

Step-by-Step: Creating HLD & LLD draw.io Diagrams for E-Commerce App (Interview Reference)

1. Preparation

- Decide on the system's main components: User Service, Product Service, Order Service, Payment Gateway, and their respective databases.
 - Identify key flows: data storage, service-to-service communication, and external integrations.
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2. Creating the HLD (High-Level Design)

2.1 Open draw.io

- Go to <https://app.diagrams.net/> or open the desktop app.
- Start a new blank diagram.

2.2 Add Components

- Drag rectangles for each service: "User Service", "Product Service", "Order Service".
- Drag cylinder shapes for each database: "User DB", "Product DB", "Order DB".
- Drag a rounded rectangle or cloud for "Payment Gateway".

2.3 Arrange Components

- Place services at the top/middle, DBs below their respective services, Payment Gateway at the bottom.

2.4 Connect Components

- Use arrows to connect:
 - User Service → User DB
 - Product Service → Product DB
 - Order Service → Order DB
 - Order Service → Payment Gateway
 - User/Product Service → Order Service (for orchestration)
- Right-click arrows to add labels (e.g., "API Call", "DB Write").

2.5 Style and Group

- Color services (e.g., green), DBs (blue), external (yellow).
- Group related items (e.g., all DBs together).
- Add a legend for color/symbol meaning.

2.6 Save/Export

- Save as .drawio file for editing.
 - Export as PNG/SVG for sharing.
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3. Creating the LLD (Low-Level Design)

3.1 Add a New Page/Layer

- In draw.io, add a new page for LLD.

3.2 Sequence Diagram (Order Placement)

- Drag UML lifeline shapes for: User, Order Service, Product Service, Payment Gateway.
- Use arrows to show:
 - User → Order Service: "POST /orders"
 - Order Service → Product Service: "GET /products/:id"
 - Order Service → Payment Gateway: "POST /pay"
 - Order Service (self): "Save Order" (dashed arrow for DB write)
- Label each arrow with the API call or action.

3.3 Data Models (Optional)

- Use UML class shapes to show data models (e.g., Order, User, Product).
- List key fields in each class.

3.4 Add Comments

- Use sticky notes or text boxes to add interview talking points (e.g., "Order Service is stateless for scaling").

3.5 Save/Export

- Save the updated .drawio file.
 - Export LLD page as needed.
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4. General Rules & Tips

- Label every component and connection.
 - Use consistent shapes/colors for clarity.
 - Arrows must indicate direction and type (solid for sync, dashed for async/error).
 - Add a legend and comments for interview reference.
 - Use layers/pages to separate HLD and LLD.
 - Practice walking through the diagram out loud as you would in an interview.
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With these steps, you can confidently create, explain, and demo HLD/LLD diagrams in any interview using draw.io!